

Part 1 of 6 · Tier 1 · 2x BTS7960 · 4x NFP-36GP-555-EN geared motors with rear hall encoders (Part 4 rev B) · Arduino Nano V3 (CH340) closes the speed loop and streams odometry to the Pi 5



- Layout is schematic; physical positions follow Part 3 sheet 4 and Part 4 rev B sheet 2.*

Spare: D2, A4, A5, A6, A7. Encoder pins are fixed (the interrupt code reads whole ports); PWM must stay on D3 / D9 / D10 / D11.

1. Motor 6-pin plug (JST XH2.54): split it at the motor. M+ / M− go to the driver on 18 AWG through the Ø12 grommets; the four encoder wires (VCC, GND, A, B) go to the Tier 1 Nano board on 4-core 24 AWG through the Ø10 grommets at (150, 65) and (150, 335).
2. Encoder VCC takes 3.3–5 V only: take it from the +5 V rail, never from 12 V. Read the wire order off the motor label or datasheet before plugging in.
3. Keep each encoder's four wires together and away from the motor leads and PWM wires. Label FL / RL / FR / RR at both ends; heat-shrink every joint.
4. The encoders have their own pull-ups, so A / B swing 0–5 V straight into the Nano. No extra resistors.
5. R_IS / L_IS stay unconnected: stalls are detected from the encoders. R_EN and L_EN are tied on each driver.
6. Do not swap M+ / M− to fix direction: the bring-up check (Part 4, sheet 8, step 3) stores the motor and encoder signs in the Nano.
7. Switch 12 V and Battery B on before (or with) the Pi, so the Pi does not back-feed the Nano, the drivers' logic and the encoders over USB.
8. Load on the Tier 1 rails ≈ 0.15 A (Nano, 4 encoders, both drivers' logic), as in sheet C.